



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 14: Business intelligence, risk management and scenarios

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed the following:

- Understanding of the fundamentals of systems thinking.
- Exploration and analysing the diversity of causal loop diagrams in systems thinking
- Demonstration of systems thinking process with the use of the Vensim software
- Adaptation of the entire concept to the semester project

Learning Outcomes for the Current Lecture

The expected learning outcomes include:

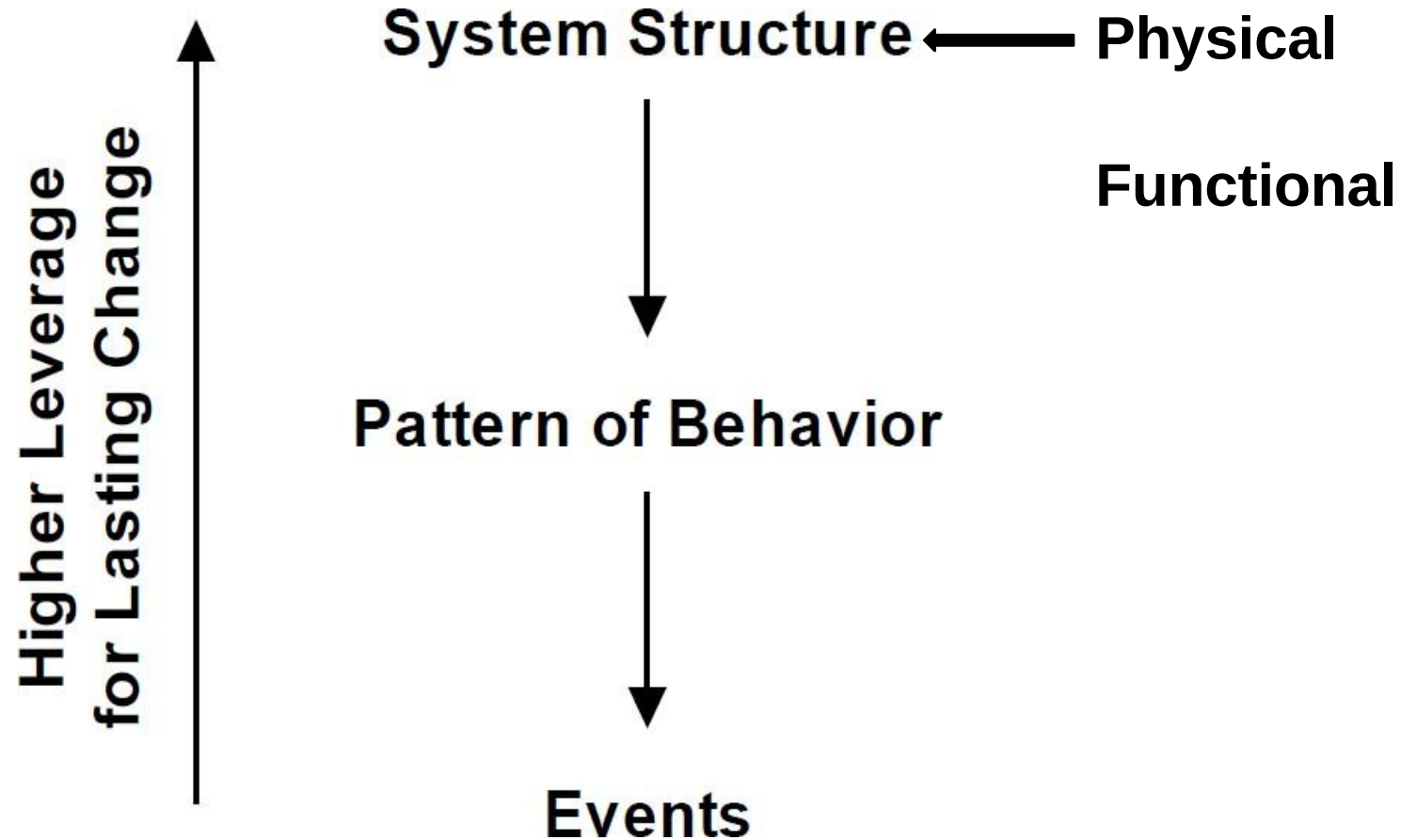
- Ability to explain, analyse, and evaluate business systems intelligence from systems thinking and sentiment analysis.
- Application and evaluation of Naïve Bayes classifier in sentiment analysis for business systems intelligence.
- Understanding, analysis, and evaluation of risks in systems and businesses.
- Ability to create scenarios for the application , analysis and evaluation of the entire concepts for the semester project.

Presentation Outline

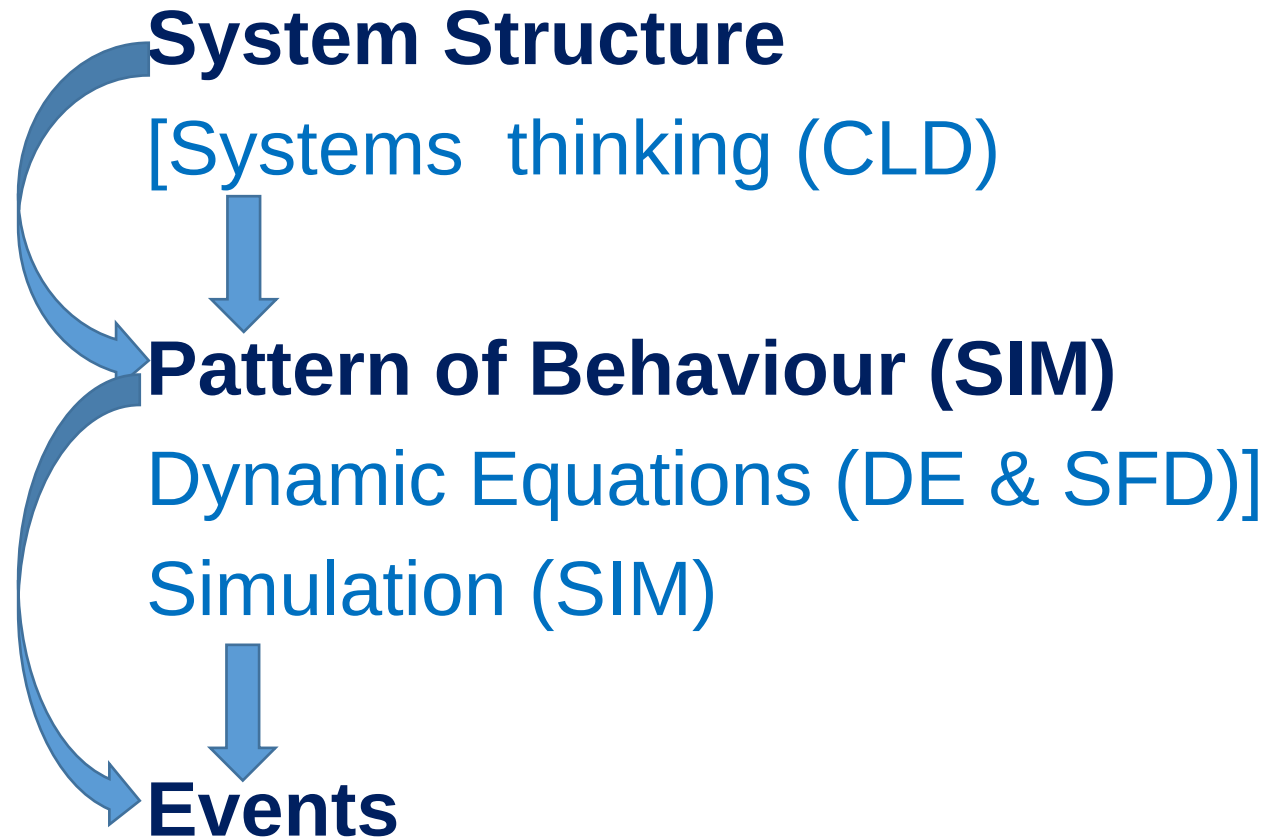
- Behavioural Intelligence from systems thinking
- Sentimental Intelligence, Bayes and Naïve Bayes Theorems
- Risk analysis and Mapping
- Adaptation of the entire knowledge to the semester project

Behavioural Intelligence

Holistic Picture of Systemic Problems Emergence



Hierarchical Levels in (Systems Thinking & Dynamics) Modelling



Legend:

CLD-Causal Loop Diagram

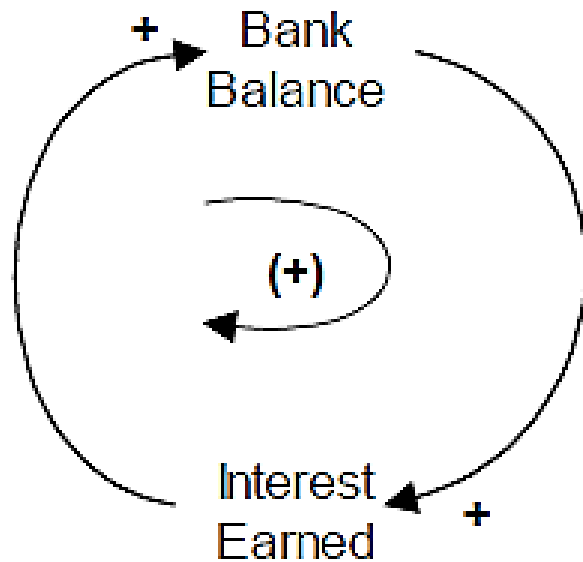
SFD-Stock & Flow Diagram

SIM- Simulations

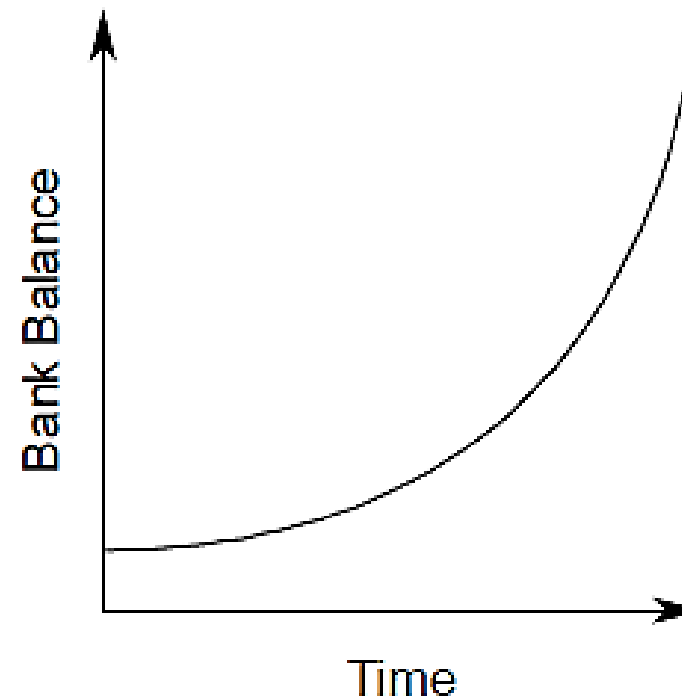
DE-Dynamic Equations

Positive (Reinforcing) Feedback Loop

System Structure



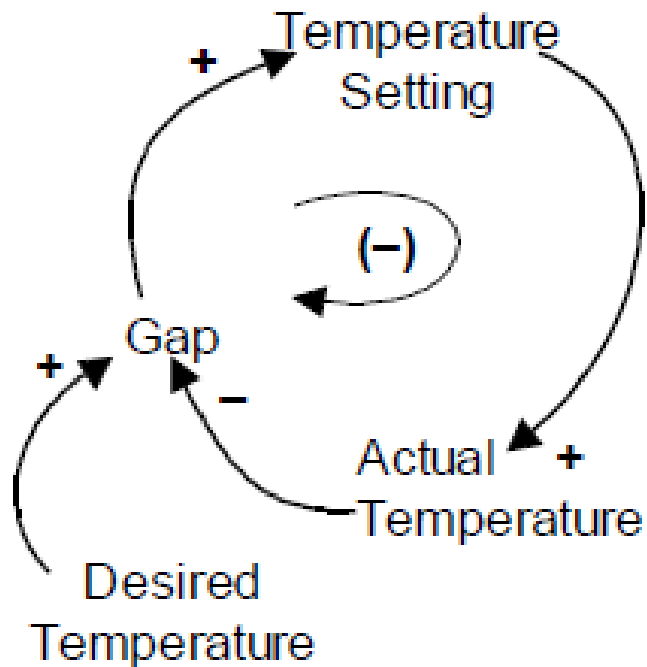
Pattern of Behavior



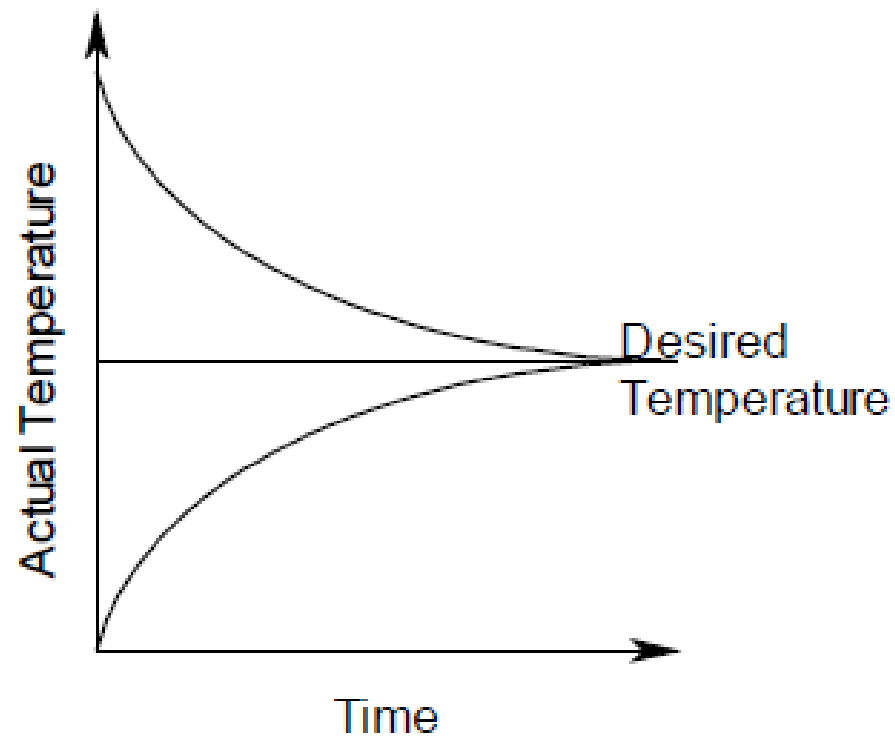
Positive (reinforcing) feedback loop: Growth of bank balance

Negative (Balancing) Feedback Loop Contd.....

System Structure



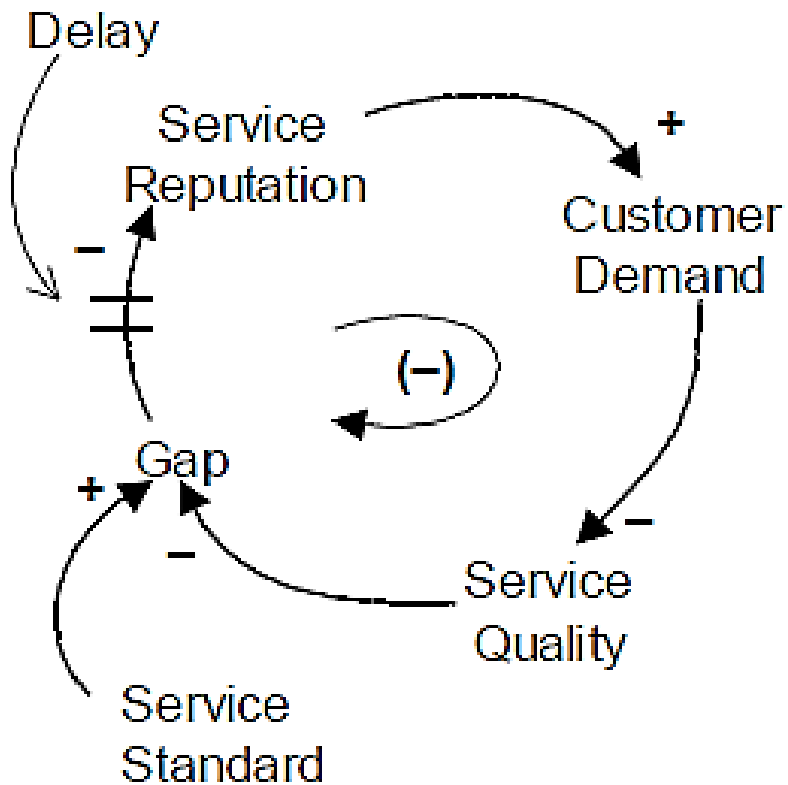
Pattern of Behavior



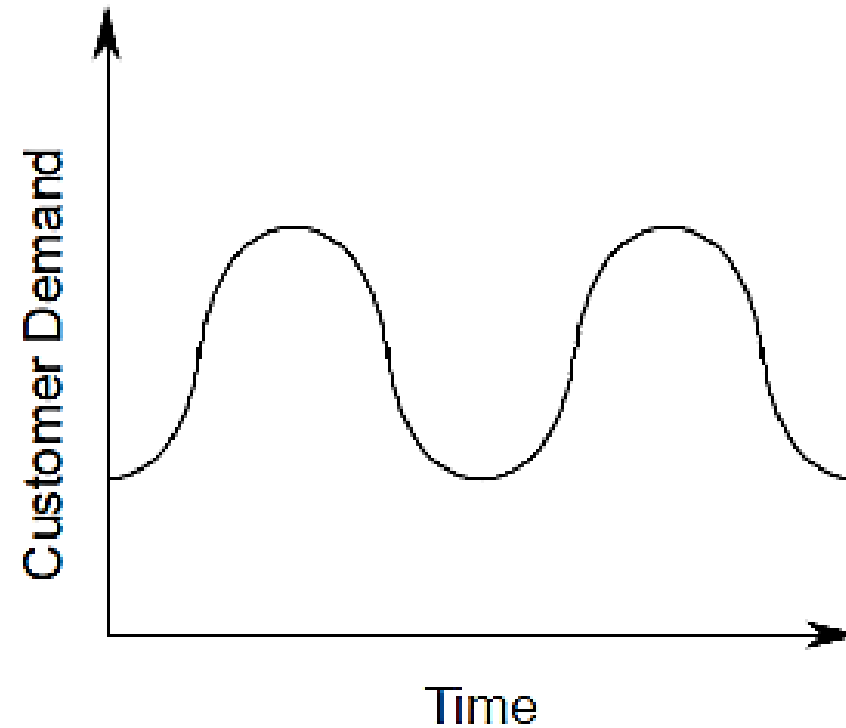
A negative feedback loop diagram for the regulation of an electric blanket temperature.

Negative Feedback Loop with Delay

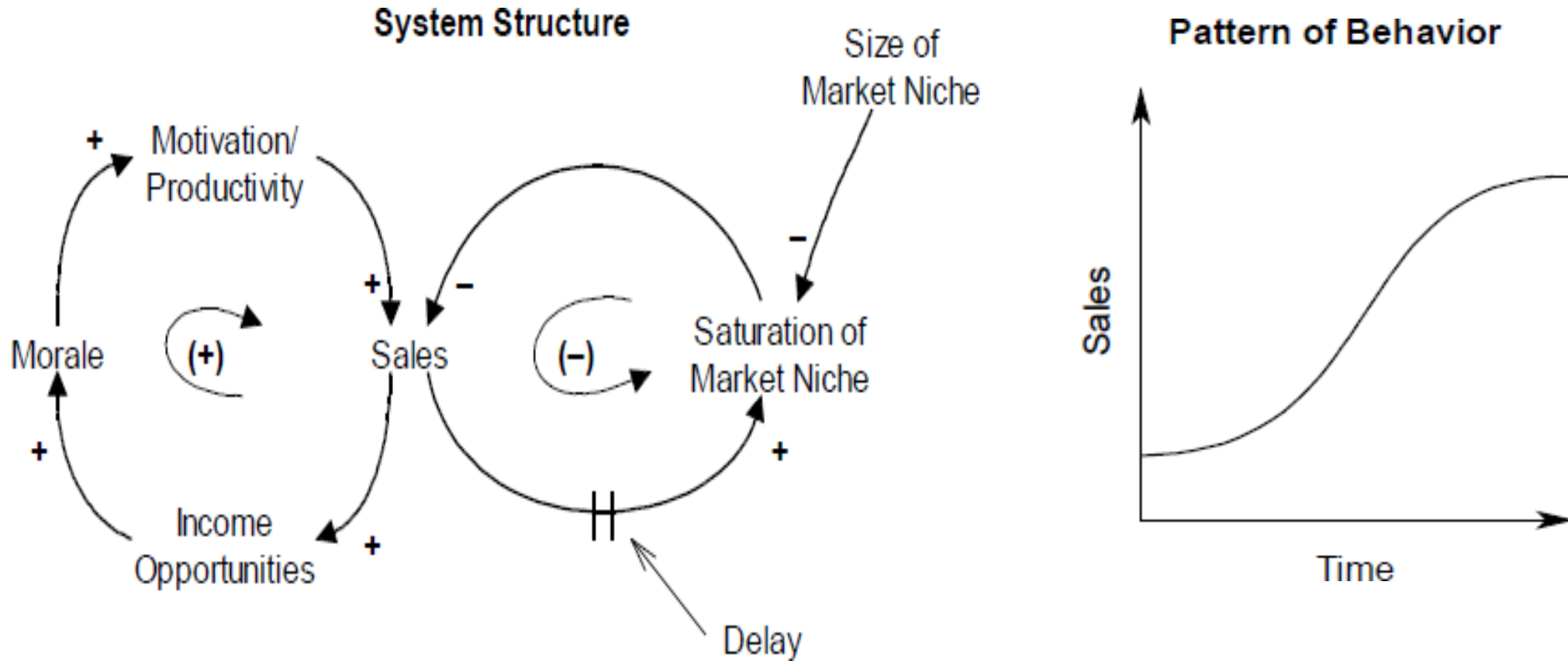
System Structure



Pattern of Behavior



Combination of Positive and Negative Loops



Sentimental Intelligence, Bayes and Naïve Bayes Theorems

Sentimental Intelligence

What is Sentiment Analysis?

Sentiment analysis, also known as opinion mining is a subfield of AI [(NLP) Natural Language Processing] that focuses on determining the emotional tone behind a body of text.

This analysis aims to identify and extract subjective information from textual data, such as attitudes, opinions, and emotions. The goal is to classify the polarity of the text, whether it is positive, negative or neutral etc.

Bayes Theorem

The expression below depicts a generalised form of the Bayes theorem.

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_i^n P(A_i)P(B | A_i)}$$

Where B is the chance event and A_i a set of mutually exclusive events for $i=[1 \text{ to } n]$.

The expression below depicts an expanded form the Bayes theorem.

$$P(A_1 | B) = \frac{P(A_1)P(B | A_1)}{P(A_1)P(B | A_1) + P(A_2)P(B | A_2) + \dots + P(A_n)P(B | A_n)}$$

Naïve Bayes Theorem

The expression below depicts the Naïve Bayes theorem.

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)}$$

$P(A|B)$ is the posterior probability i.e. the Probability of a Hypothesis A, given that event B occurs.

$P(B|A)$ is the likelihood probability i.e. the probability of the evidence given that hypothesis A is true.

$P(A)$ is prior probability, i.e. the probability of the hypothesis before observing the evidence, and

$P(B)$ is marginal i.e. the probability of the evidence.



Application of Bayes Theorem

Scenario #1:

30% of the total output of a factory is produced in workshop A while the other 70% is produced in workshop B. Also, 15 out of every 1000 components produced in A are defective and 8 out of every 1000 component produced in B are defective. If a component drawn at random from a mixture of the output of A and B is found to be defective, what is the probability that it is from workshop A?

Application of Bayes Theorem Contd...

Solution

The probability that a component comes from A is $P(A) = \frac{30}{100} = 0.3$. The probability that it is defective given that it comes from A is $P(D|A) = \frac{15}{1000} = 0.015$.

Similarly $P(B) = 70\% = \frac{70}{100} = 0.7$; $P(D|B) = \frac{8}{1000} = 0.008$.

Using Bayes theorem, the probability that the component comes from A and is defective is given as:

$$P(A|D) = \frac{P(A).P(D|A)}{P(A).P(D|A) + P(B).P(D|B)} = \frac{0.3 * 0.015}{(0.3 * 0.015 + 0.7 * 0.008)} = 0.45$$

Application of Naïve Bayes Theorem

Scenario #2

A Rugby match slated for a predefined day is being debated upon for cancellation due to reasons bothering around possible unfavourable weather conditions. Based on varying sentiments, you are required to calculate the probability of the players, playing during this day.

Application of Naïve Bayes Theorem

Steps towards addressing the above problem:

- i. Convert the data set to a frequency table
- ii. Create a likelihood table by finding the probabilities like overcast probability
- iii. Use the Naïve Bayesian equation to calculate the posterior probability for each class.

Application of Naïve Bayes Theorem

Contd...

Solution #2: Sample data from different forecasting media have been analysed below

Weather	Play
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes
Sunny	Yes
Overcast	Yes
Rainy	No
Rainy	No
Sunny	Yes
Rainy	Yes
Sunny	No
Overcast	Yes
Overcast	Yes
Rainy	No

Frequency Table		
Weather	No	Yes
Overcast		4
Rainy	3	2
Sunny	2	3
Grand Total	5	9

Likelihood table				
Weather	No	Yes		
Overcast		4	=4/14	0.29
Rainy	3	2	=5/14	0.36
Sunny	2	3	=5/14	0.36
All	5	9		
	=5/14	=9/14		
	0.36	0.64		

Application of Naïve Bayes Theorem

Contd...

Sunny Condition

$$P(\text{Sunny}|\text{Play}) = 3/9 = 0.33$$

$$P(\text{Play}) = 9/14 = 0.64$$

$$P(\text{Sunny}) = 5/14$$

$$P(\text{Play}|\text{Sunny}) = [P(\text{Sunny}|\text{Play}) * P(\text{Play})] / P(\text{Sunny})$$

$$= [(0.33 * 0.64) / 0.36] = 0.59$$

Application of Naïve Bayes Theorem

Contd...

Overcast Condition

$$[P(\text{Overcast}|\text{Play})=4/9 = 0.44$$

$$P(\text{Play}) = 9/14 = 0.64$$

$$P(\text{Overcast}) = 4/14 = 0.29$$

$$\begin{aligned} P(\text{Play}|\text{Overcast}) &= [P(\text{Overcast}|\text{Play}) * P(\text{Play})]/P(\text{Overcast}) \\ &= [(0.44 * 0.64)/0.29] = 0.97 \end{aligned}$$

Application of Naïve Bayes Theorem

Contd...

Rainy Condition

$$[P(\text{Rainy}|\text{Play})=2/9 = 0.22$$

$$P(\text{Play}) = 9/14 = 0.64$$

$$P(\text{Rainy}) = 5/14 = 0.36$$

$$P(\text{Play}|\text{Rainy}) = [P(\text{Rainy}|\text{Play}) * P(\text{Play})] / P(\text{Rainy})$$

$$= [(0.22 * 0.64) / 0.36] = 0.39$$

Risk Analysis & Mapping

Basics on Risk Management

What is Risk?

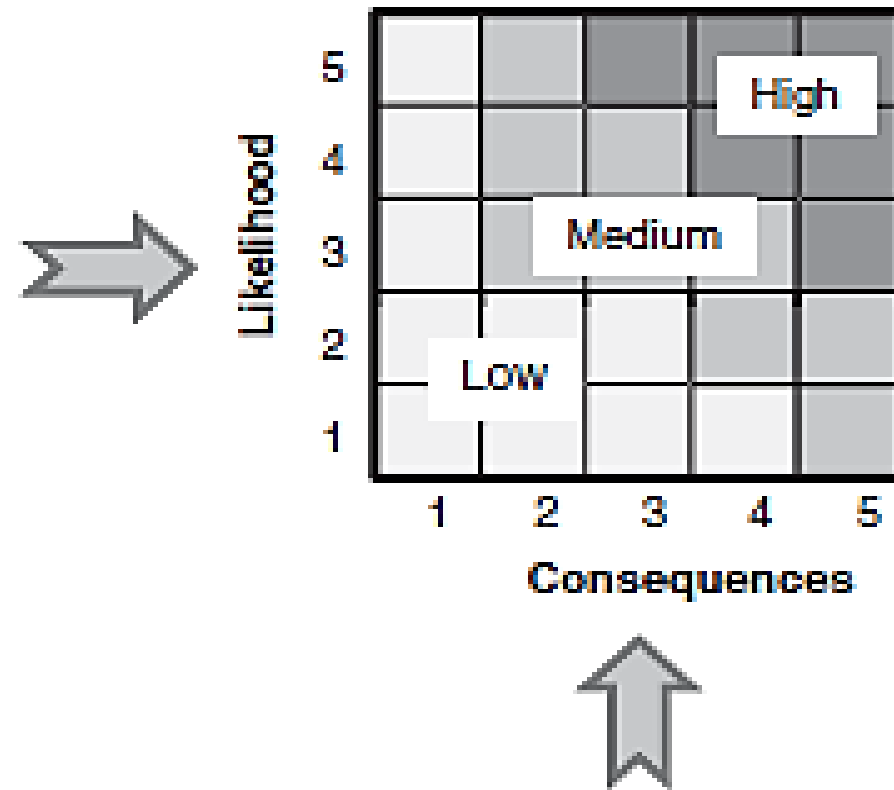
- ❖ It's a measure of negativity associated with an action
- ❖ It's a measure of hazard or a situation that involves exposure to danger
- ❖ It is an unpredictable or a predictable negative situation that arises from a prior action

The act of identifying and minimising risk in business systems design and development is called risk management .

Components of Risk Management

- **The Risk Management Guide divides the subject of risk management into:**
 - ❖ risk planning,
 - ❖ risk prioritization,
 - ❖ risk handling, and risk monitoring.
- **However these can be combined into two categories:**
 - ❖ Risk assessment (risk planning and prioritization)
 - ❖ Risk mitigation (risk handling and monitoring).

Risk Chart



Risk Likelihood and Consequence chart

What is the likelihood the risk will happen?		
Level		Your approach and processes ...
Likelihood	1 Not likely	... Will effectively avoid or mitigate this risk based on standard practices
	2 Low likelihood	... Have usually mitigated this type of risk with minimal oversight in similar cases
	3 Likely	... May mitigate this risk, but work-arounds will be required
	4 Highly likely	... Cannot mitigate this risk, but a different approach might
	5 Near certainty	... Cannot mitigate this type of risk; no known processes or work-arounds are available

Risk Likelihood

Risk likelihood	Design status
High	• Significant extension from past designs • Multiple new and untried components • Complex components and/or interfaces • Marginal analytical tools and data
Medium	• Moderate extension from past designs • Components complex but not highly stressed • Analytical tools available
Low	• Application of qualified components • Components of medium complexity • Mature technologies and tools

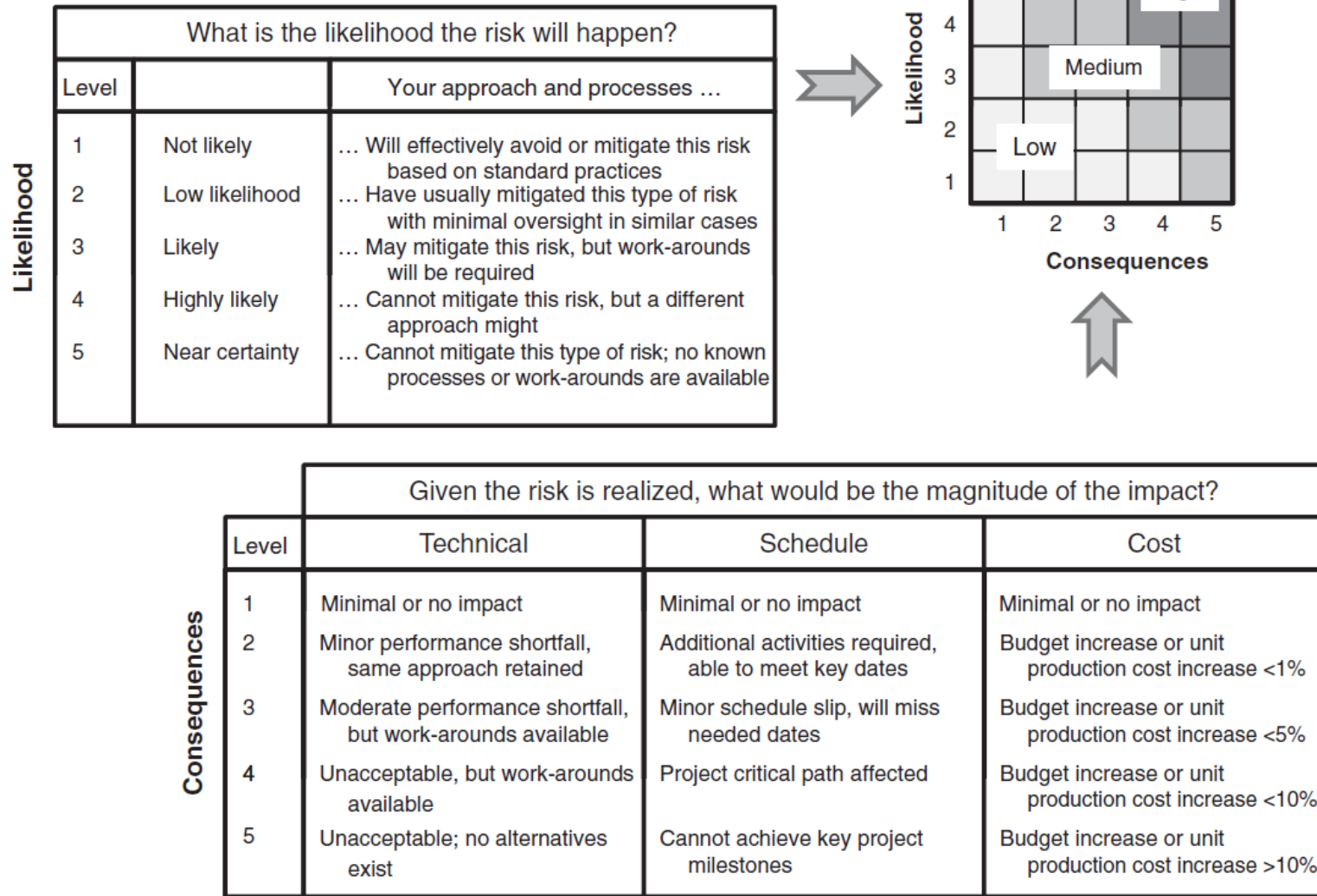
Risk Consequence chart

		Given the risk is realized, what would be the magnitude of the impact?		
Consequences	Level	Technical	Schedule	Cost
	1	Minimal or no impact	Minimal or no impact	Minimal or no impact
	2	Minor performance shortfall, same approach retained	Additional activities required, able to meet key dates	Budget increase or unit production cost increase <1%
	3	Moderate performance shortfall, but work-arounds available	Minor schedule slip, will miss needed dates	Budget increase or unit production cost increase <5%
	4	Unacceptable, but work-arounds available	Project critical path affected	Budget increase or unit production cost increase <10%
	5	Unacceptable; no alternatives exist	Cannot achieve key project milestones	Budget increase or unit production cost increase >10%

Risk Criticality

Criticality	System impact	Program impact
High	• Major degradation in performance (50–90%) • Serious safety problem	• Major increase in cost and/or schedule (30–70%) • Production cutbacks
Medium	• Significant degradation in performance (10–50%) • Short losses of operability • Costly operational support	• Significant increases in cost and/or schedule (10–30%) • Intense reviews and oversight • Production delays
Low	• Minor degradation in performance (>10%) • Occasional brief delays • Increased maintenance	• Minor increase in cost and/or schedule (<10%) • Vigorous reviews and oversight

Derived from
Risk Management—A Process Overview by Bob Skalamera



Risk Likelihood and Consequence chart

Source: Systems Engineering Principles and Practice by Kossiakoff, A. et al.

Sample Risk Plan Worksheet

Risk title:

Project name:

Risk owner:

Last updated:

Team:

Date submitted:

Description of risk:

Risk type:

☐ Technical

☐ Schedule

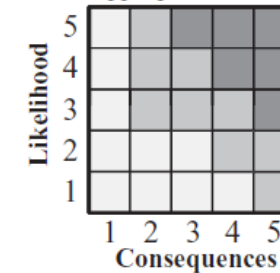
☐ Cost

☐ Other

Statement of basic cause:

Consequence if risk is realized:

Place X, 1, 2, ... in the appropriate cells.



Risk reduction plan

Risk level
if successful

Action/milestone event.

Date

Success criteria

L

C

Comments

1.

2.

3.

4.

Adaptation to the Semester Project

- **Section 5 (PART A) (BSDM)**
 - Present a consolidated risk matrix with a brief description of a mitigation strategy for each
 - Business intelligence

Learning Outcomes Evaluated (Key Takeaways)

- From this lecture, you should be able to view a systems thinking loop and have an estimated mindset of the expected behaviour of the loop based on the structural interaction setup by the designer of the system.
- Furthermore, from this lecture, you should be able to gather sentiment data about one or more critical goals in your business and deploy the Naïve Bayes classifier theorem in a bid to plan ahead i.e., business intelligence.
- You should be able to identify risks pertinent to your business, categorise these on the risk chart (mapping) based on their levels of likelihood and potential consequence.
- **Can you attempt the above, as expected of you?**

For further reading: Section covered in e-book, Hardcover text and others include:

- Source: **Part 1, Section 5-** Systems Engineering Principles and Practice by Kossiakoff, A. et al.
- Artificial Intelligence by Pankaj Sharma: Chapter 4 Machining Learning
- Other Alternative Sources

The End

Next Topic

Unit 15: Project Management KPIs



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